VITHIYA V

Fresher [B.Tech CSE, Puducherry Technological University]

Marakkanam, Tamil Nadu | 8344455990 | vvithiya537@gmail.com | **LinkedIn:**linkedin.com/in/vithiya-velu-306363314 | **GitHub:** github.com/VithiyaVelu

PROFILE

Computer Science Engineering student (CGPA: 8.3/10) with hands-on experience in software development, Python, AI/ML, and database technologies through internships and projects. Strong foundation in problem-solving, data structures, and application development. Eager to contribute technical skills to software and IT engineering roles.

EDUCATION

B.Tech in Computer Science & Engineering | Puducherry Technological University | 2022 – 2026 | CGPA: 8.3/10

HSC | The Study School, Kalapet | 2022 | 68%

SSLC | The Study School, Kalapet | 2020 | 82.5%

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, Dart, SQL, C#
- **Frameworks:** Django, Flask, Spring, Flutter, RESTful APIs
- **AI / ML:** XGBoost, LightGBM, CatBoost, Scikit-Learn, Q-Learning
- **Databases:** MySQL, SQL Server, SQLite
- **Web & Tools:** HTML,CSS JavaScript, AJAX, Git, Docker
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

INTERNSHIPS

App Developer Intern | Cornerstone Engineering Solutions, Puducherry | *Dec 2025 – Jan 2026*

- Built a Smart Expense Tracker Android app using Flutter & Dart with real-time balance tracking, state management, and full CRUD operations.

Data Science & ML Intern | HDLC Technologies, Chennai | *Jun 2025 – Jul 2025*

- Engineered a California House Price Prediction model using Linear Regression & Random Forest via Scikit-learn; executed data preprocessing, feature engineering, and metric validation.

Data Science Intern / AiQsoftTech, Chennai / *May 2024 – Jun*

- Worked on a machine learning project to predict the likelihood of railway ticket booking using historical ticketing data.

PROJECTS

Smart Fitness Checker & Multi-Paradigm Intelligent System — *Python, Flask, Docker, Scikit-Learn, Prolog, Q-Learning, A*/AO**

- Architected a microservices-based health recommendation dashboard deployed via Docker with expert system (Prolog), A*/AO* graph search, Decision Tree, K-Means, and Q-Learning modules.

Industrial Steel Defect Detection & Risk Analysis — *XGBoost, LightGBM, CatBoost, ADASYN, PCA, Streamlit | Tata Steel AI Hackathon*

- Built an end-to-end AI quality control pipeline using ensemble ML, PCA for dimensionality reduction, ADASYN for class imbalance, and a Streamlit dashboard for real-time defect analytics.

Django Social Media Analytics Dashboard — *Django, SQLite, Scikit-Learn, vaderSentiment, TextBlob, gpt4all | Skillrank Hackathon*

- Designed a high-volume analytics pipeline processing 500,000+ posts with TF-IDF/KMeans clustering, dual-layer sentiment engine, and a local LLM (orca-mini-3b) chat assistant.

Real-Time Farming Support Mobile App — *Java, Android Studio, RESTful APIs*

- Developed a native Android app with live crop updates, finance tracking, and disease alerts using advanced OOP, multithreading, and custom REST API integration.

CERTIFICATIONS & ACHIEVEMENTS

- NIELIT Calicut (MeitY, Govt. of India) — Machine Learning using Python, Feb 2026
- Hackathon: Tata Steel AI Hackathon — Production-ready ML defect detection pipeline
- Hackathon: Skillrank Hackathon at PTU — End-to-end sentiment engine on 500k data nodes
- Workshops: Full Stack Dev (Capgemini) | Kubernetes & AI-LLM (PTU & Nokia) | IoT Applications (IEI India)
- NPTEL: Cognitive Psychology (2023–24) | Wildlife Ecology (2024–25) | Python @ Bita Academy (2 months)